Introduction to Economic Principles for Engineering

What is Economics

- Study of how people and organizations allocate scarce resources

- ***Economics*** **– the study of how to allocate/distribute scarce resources effectively among alternative ends/wants**

Importance of Economics in engineering

- Helps in making strategic business and project decisions/planning.

- Understanding cost, value, and benefit in software development.

- Evaluating demand, supply, and pricing for tech products/software.

- Managing resources allocation in IT projects efficiently.

IT services, and software products must be developed, priced, and sold in competitive markets where cost value, efficiency , and profit matter.

What are Economic Principles?

- Economic principles are basic concepts and analytical tools such as demand theory, production theory, cost theory, used to understand decision-making.

- these principles are applied to software, IT services, and tech-based business models.

Engineering Economics

- Application of economic principles, concepts, and techniques to engineering and technology-related decisions.
- Evaluate the economic feasibility of engineering projects and decisions

- For example It involves evaluating the costs and benefits of engineering projects, making choices between alternatives, and determining the most economically viable option for resource allocation, design, production, or investment.

Engineering economics helps engineers:

- To choose between alternatives
- minimise costs, maximising the benefits of the projects
- Ensure that engineering poducts are not only technically sound but also economically viable

Micro vs Macro Economics

- Microeconomics: Individual decision-makers like consumers and firms.
- Macroeconomics: National and global economic performance.

For example: Microeconomics is like studying the performance of a single app or user while Macroeconomic is like studying the entire digital ecosystem or economy.

Microeconomics helps to Set software pricing, Minimize project cost and understand consumer preferences

Where as Macroeconomics helps to Predict IT job market behaviour .

Make better investment or entrepreneurship decisions and track market trends (e.g., recession, boom)

Scope of Economics for Engineers

- **Understanding Resource Allocation:**

Economics helps to analyse how limited resources (time, money, manpower) can be used optimally. Economic principles guide rational decision-making when resources, time, or options are limited. Learn to manage limited tech and human resources efficiently.

Helps in planning project budgets, team salaries, and cloud/storage expenses

- **Cost Analysis**

Understand the different types of costs. **Examples: Choosing between two software development tools. Selecting the most cost-effective material for a machine component.**

Pricing Strategies for Tech Products

Apply concepts like demand and marginal cost for setting pricing SaaS or apps.

- **Investment Decision-Making in Technology:**

Use economic principles to evaluate IT investments projects. Decide whether to invest in cloud, servers, AI tools, or cybersecurity upgrades.

key economic problems: scarcity and resource allocation.

- Unlimited wants: Human desires and needs are never-ending or unlimited. As one need is satisfied, new wants arise, often driven by technology, lifestyle, or economic development.

- Scarcity refers to the limited availability of resources (like land, labour, capital, and raw materials) to meet the unlimited human wants.

Economic Problem – the problem of having unlimited wants, but limited resources to satisfy them.

we do not have enough resources to produce all the things (for example software) we would like to have. Cloud computing resources like server space and bandwidth are limited. Developers must optimize applications accordingly.

Resources Allocation

The process of deciding how scarce resources are distributed to produce different goods and services efficiently. Goal: Maximize productivity, reduce waste, and meet consumer needs.

Deciding which feature to prioritize in a software update with limited development time.

Developers and IT managers must allocate processing power, time, and code efficiency wisely. |

Engineers must optimize use of materials, time, and technology in projects.

- Resource allocation is about choosing the best possible useof available resources.
- this help engineers and tech professionals design solutions that are efficient, and cost-effective

- The fundamental economic problem is Scarcity. It leads three basic economics problems:

1. what to produce: the problem of selection of goods

2. How to produce: the problem of selection of production method

3. For whom to produce: selection of category of customers who will ultimately purchase your product.

what to produce

The problem of what to produce requires choosing which products or services to develop with limited resources.

For example App Developers:

Should build a food delivery app, or a college portal?

Decision depends on market gap, user base, and monetization. You must evaluate demand, utility, and technical feasibility before deciding what software or service to build.

How to produce

- This problem focuses on choosing the most efficient method or technology to produce goods and services. It is about choosing the best production method to complete a project efficiently within resource limits.

Example:

- Should you develop software using open-source tools (e.g., Python, VS Code) or premium IDEs and platforms

Decision depends on:

- Budget
- Skill level
- Project requirements

For whom to produce

- identifying the target customers or user group for a product or service. Who has also the buying capacity or ability to buy it.
- you should know that that not all software is for everyone.

Knowing who will use a product and can buy your product helps in customizing its design, features, cost, and marketing.

- For whom to produce tells the importance of identifying and prioritizing the right user.
- In software development, this ensures that the right product reaches the right audience, achieving both efficiency and satisfaction.